Points: 17/17 (100%) well done!

Algebraic number theory
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Serie 3

Ex 1.)

Let R be a ring and let $I, J \subseteq R$ be ideals.

t.s. i) $I \cap J \subseteq R$ is an ideal, ii) $I + J \subseteq R$ is an ideal.
iii) $IJ \subseteq R$ is an ideal.
iv) It holds that $IJ \subseteq I \cap J$ and $I \cup J \subseteq I + J$.
v) $IJ = I \cap J$ does not hold in general.

proof i) Since $I, J < R$ are subgroups, $I \cap J < R$ is as well ($0 \in I, J$, $\forall a, b \in I, J \Rightarrow a - b \in I, J$)
Let $a \in I \cap J$ and $r \in R$, then $a \in I, a \in J$ thus $ra \in I$, $ra \in J$ thus $ra \in I \cap J$
since $I, J \subseteq R$ are ideals by assumption. (2)

ii) Since $0 \in I, J \Rightarrow 0 \in I + J$
Let $a, b \in I + J$. Then there exist $a_1, b_1 \in I$ and $a_2, b_2 \in J$ such that

$$a = a_1 + a_2, \quad b = b_1 + b_2$$

Then it holds that

$$a - b = \underbrace{(a_1 - b_1)}_{\in I} + \underbrace{(a_2 - b_2)}_{\in J} \in I + J$$

Thus $I + J < R$ is a subgroup of R (w.r.t. $+$)
Then let $r \in R$ and it holds

$$ra = r(a_1 + a_2) \overset{\text{since } a_1, a_2, r \in R}{=} \underbrace{ra_1}_{\in I \text{ by def}} + \underbrace{ra_2}_{\in J \text{ by def}} \in I + J$$

Thus $I + J \subseteq R$ is an ideal. (2)

iii) Since $0 \in I, J \Rightarrow 0 \in I \cdot J$.
Let $a, c \in I \cdot J$ and $r \in R$. Then there exist $n, m \geq 0$, $\begin{cases} a_1, \dots, a_n \in I, b_1, \dots, b_n \in J \\ c_1, \dots, c_m \in I, d_1, \dots, d_m \in J \end{cases}$

such that

$$a = a_1 b_1 + \dots + a_n b_n, \quad b = c_1 d_1 + \dots + c_m d_m$$

Then it holds that

$$a - b = a_1 b_1 + \dots + a_n b_n + (-c_1) d_1 + \dots + (-c_m) d_m, \text{ where } a_i, -c_i \in I, b_i, d_i \in J \text{ and } n + m < \infty$$

thus $a - b \in IJ$ and therefore $IJ < R$ a subgroup.
Furthermore, it holds that

$$ra = r(a_1 b_1 + \dots + a_n b_n) = \underbrace{ra_1}_{\in I} b_1 + \dots + \underbrace{ra_n}_{\in I \text{ by def}} b_n \in IJ$$

Thus $ra \in IJ$, thus $IJ \subseteq R$ is an ideal. (2)

iv) Let $a \in IJ$. Then there exist $n \geq 0$ and $a_1, \dots, a_n \in I$, $b_1, \dots, b_n \in J$ s.t. $a = a_1 b_1 + \dots + a_n b_n$.
By i), $I \cap J$ is an ideal thus for each $i \in \{1, \dots, n\}$, we get $a_i b_i \in I \cap J$,
since $a_i \in I \subseteq R$ and $b_i \in J \subseteq I \cap J$, furthermore it implies that

$$a = a_1 b_1 + \dots + a_n b_n \in I \cap J \quad (I \cap J < R \text{ additive group})$$

This implies $IJ \subseteq I \cap J$. Let $b \in I \cup J$. Then if $b \in I$ we get by $0 \in J$ that $b = b + 0 \in I + J$ and if $b \in J$ we get by $0 \in I$ that $b = 0 + b \in I + J$
thus $I \cup J \subseteq I + J$. (2)

v) We consider the case $R = \mathbb{Z}$, $I = 2\mathbb{Z}$, $J = 4\mathbb{Z}$.
Then $I \cap J = 4\mathbb{Z}$ since $4\mathbb{Z} \subseteq 2\mathbb{Z}$, thus $4 \in I \cap J$.
Assuming $4 \in IJ$, there exist finitely many $a_i, b_i \in \mathbb{Z}$ s.t.

$$4 = \sum_i 4 a_i 2 \cdot b_i$$

$$= 8 \cdot \underbrace{\sum_i a_i b_i}_{\in \mathbb{Z}}$$

which implies $8 \mid 4$ in $\mathbb{Z}$. But this is absurd since $8 > 4$
thus 4 can not be an element of IJ and we can conclude. ∎

Ex 2.)

Let R be a ring, let $I, J, K \subseteq R$ be ideals.

t.s. $I(J+K) = (IJ) + (IK)$.

proof. It holds that

$$s \in I(J+K) \overset{\checkmark}{\Leftrightarrow} s = \sum_i a_i(b_i + k_i) \text{ for finitely many } a_i \in I, b_i \in J, k_i \in K$$

$$\overset{\checkmark}{\Leftrightarrow} s = \sum_i a_i b_i + a_i k_i \text{ for } \text{---}''\text{---}$$

$$\overset{\checkmark}{\Leftrightarrow} s = \underbrace{\sum_i a_i b_i}_{\in IJ} + \underbrace{\sum_i a_i k_i}_{\in IK} \text{ for } \text{---}''\text{---}$$

$$\overset{\checkmark}{\Leftrightarrow} s \in (IJ) + (IK).$$

Thus the inclusions $I(J+K) \subseteq (IJ) + (IK)$, $I(J+K) \supseteq (IJ) + (IK)$ holds and therefore its equality.

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Ex 3.)

Let $d > 1$ be an integer.

t.s. $\mathbb{Z}[\sqrt{-d}]^\times = \{-1, 1\}$.

proof. The inclusion $\{-1, 1\} \subseteq \mathbb{Z}[\sqrt{-d}]^\times$ holds since $\{-1, 1\} = \mathbb{Z}^\times \subseteq \mathbb{Z}[\sqrt{-d}]^\times$. ✓

On the contrary let $u \in \mathbb{Z}[\sqrt{-d}]^\times$.

Then there exist $a, b, e, f \in \mathbb{Z}$ such that

$u = a + b\sqrt{-d}$ and $u \cdot u^{-1} = 1$ where $u^{-1} = e + f\sqrt{-d}$. ✓

Thus we get

$$1 = N(u \cdot u^{-1}) \overset{\text{Ex 1.}}{=} N(u) \cdot N(u^{-1})$$
$$= \underbrace{(a^2 + b^2 d)}_{\in \mathbb{Z}_{\geq 0}} \underbrace{(e^2 + f^2 d)}_{\in \mathbb{Z}_{\geq 0}} \text{ since } d > 1. \quad ✓$$

This implies $N(u) = 1$ since $\mathbb{Z}^\times = \{-1, 1\}$ and $N(u) \geq 0$. ✓

Hence,

$$1 = N(u) = a^2 + b^2 d.$$

Since $d > 1$, $a^2 \geq 0$ thus $b^2 d \leq 1$ we get $b^2 < 1$ and therefore $b^2 = 0$ ($b \in \mathbb{Z}$). Since $\mathbb{Z}$ an ID this provides us $b = 0$. ✓

Then

$$1 = N(u) = a^2, \text{ thus } a \in \{-1, 1\} = \mathbb{Z}^\times. \quad ✓$$

Hence $u = a \in \{-1, 1\}$ and the other inclusion implies and therefore the equality.

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Ex 4.)

Let K, L be numberfields and let $\varphi: K \longrightarrow L$ be a ringhom.

t.s. $\varphi(\mathbb{Z}_K) \subseteq \mathbb{Z}_L$.

proof. By definition, $\mathbb{Q} \hookrightarrow K$, $\mathbb{Q} \hookrightarrow L$ are finite field extensions.

Let $x \in \mathbb{Z}_K$. Then $x \in K$ and there exists a monic $P = \sum_{i=1}^{n} a_i T^i \in \mathbb{Z}[T]$, such that $P(x) = 0$.

claim $\varphi|_{\mathbb{Z}} = id_{\mathbb{Z}}$

Then we consider the following map induced by φ:

$$\iota: \mathbb{Z}[T] \longrightarrow \mathbb{Z}[T], \sum_{i=1}^{n} a_i T^i \longmapsto \sum_{i=1}^{n} \varphi(a_i) T^i$$

Then, since $\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow K$ and by claim, ι is well defined.

Now, since $\varphi(0) = 0$ and $P(x) = 0$ we get

$$0 = \varphi(P(x)) \overset{\text{since } \varphi \text{ a RH, thus additivity, multiplicat. holds.}}{=} \sum_{i=1}^{n} \varphi(a_i) \underbrace{\varphi(x^i)}_{=(\varphi(x))^i} = \iota(P)(\varphi(x)).$$

This implies that $\varphi(x) \in L$ is a root of $\iota(P) \in \mathbb{Z}[T]$ and furthermore $\iota(P)$ is monic, since $\varphi(1) = 1$ and P monic (so $\varphi(a_n) = 1$)

Therefore $\varphi(x) \in \mathbb{Z}_L$ and the statement follows.

It is left to prove the claim.

Since φ is a ringhom, it holds that $\varphi(0) = 0$, $\varphi(1) = 1$ and $\varphi(-1) = -1$.

Then for any $z \in \mathbb{Z} \setminus \{0\}$ it holds that

$$z = \begin{cases} \underbrace{1 + \cdots + 1}_{z} & \text{for } z > 0 \\ \underbrace{(-1) + \cdots + (-1)}_{|z|} & z < 0 \end{cases}$$

This provides us for any $z \in \mathbb{Z} \setminus \{0\}$ that

$$\varphi(z) = \begin{cases} \varphi(\underbrace{1 + \cdots + 1}_{z \text{ times}}) = \underbrace{\varphi(1) + \cdots + \varphi(1)}_{z \text{ times}} = z \cdot 1 = z & z > 0 \\ \varphi(\underbrace{(-1) + \cdots + (-1)}_{|z|}) = \underbrace{\varphi(-1) + \cdots + \varphi(-1)}_{|z|} = |z|(-1) = z & z < 0 \end{cases} \quad \text{for}$$

$$= z$$

Hence, $\varphi|_{\mathbb{Z}} = id_{\mathbb{Z}}$ and we can conclude. ✓

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Bonus exercise

i) No, $I \cup J$ is not an ideal in general.

Consider $R = \mathbb{Z}$, $I = (2)$, $J = (4)$.

Then $2 \in (2) \subseteq (2) \cup (4)$ and $4 \in (4) \subseteq (2) \cup (4)$.

But $2 + 4 = 6 \notin (2)$ and $6 \notin (4)$ thus $6 \notin I \cup J$.

Hence $I \cup J \subsetneq R$ is not an additive subgroup and in particular not an ideal.

ii) Let $I, J \subsetneq R$ be ideals of a ring R, such that $IJ = R$.

t.s. $I = J = R$.

proof. Since $IJ \subseteq I \cap J \subseteq I \cup J \subseteq I + J$, it holds that $I + J = R$

Then the following isomorphism

$$\{0\} \simeq R/(I+J) \simeq (R/I)/\pi(J) \quad \text{, where } \pi : R \to R/I \text{ the canonical surjection.}$$

$I + J = R$

implies that $R/I = \pi(J) = \{r + I \mid r \in J\}$.

Therefore we get $J = R$. By replacing I with J, we get $I = R$ as well.

Alternatively this can be show by chinese reminder theorem

The following map

$$\pi : R \longrightarrow R/I \times R/J, \quad r \longmapsto (\pi_1(r), \pi_2(r)), \quad \pi_i : R \to R/I \text{ canonic. surj.}$$

is surjective if and only if $I + J = R$.

where $\ker(\pi) = I \cap J \supseteq R$. $\Rightarrow I \cap J = R$

Since $I + J = R$, we get by isomorphic theorem that

$$\{0\} \simeq R/(I \cap J) \overset{\pi}{\simeq} R/I \times R/J$$

Thus $R/I = \{0\}$, $R/I = \{0\}$ thus $I = J = R$.